

Revised Generation Interconnection System Impact Study Report

For Queue Project Q13-0264

Wekiwa 100 kV
July 31, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0264. The Interconnection Customer proposes a 220.2 MW solar photovoltaic facility interconnecting at the Wekiwa 100 kV substation in the Indiana area. The request was received on August 26, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 154 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$36,821,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: WEKIWA / W FORD FLAT SUBSTATION 100 KV
Contingency: RED LION / WEKIWA 100 KV
Loading: 157.5% (Rating: 100 MVA)
DFAX: 100.0%
Pre-project loading: 63.0%
2. Facility: COSO 7 / WEST ASHEVILLE 100 KV L
Contingency: E METHUEN / HENRY COUNTY 500 KV
Loading: 124.0% (Rating: 125 MVA)
DFAX: 10.6%
Pre-project loading: 105.3%
3. Facility: WEST ASHEVILLE / WARTBURG 100 KV L
Contingency: W ASHEVILLE / N TIFTON 100 KV
Loading: 116.2% (Rating: 144 MVA)
DFAX: 10.6%
Pre-project loading: 99.8%
4. Facility: RED LION / WEKIWA 100 KV
Contingency: WEKIWA / WEST FORD FLAT 100 KV
Loading: 115.3% (Rating: 136 MVA)
DFAX: 100.0%
Pre-project loading: 46.1%
5. Facility: SHERMAN ST / EASTMONT 100 KV
Contingency: HENRY COUNTY / EAST METHUEN 500 KV
Loading: 113.8% (Rating: 116 MVA)
DFAX: 7.4%
Pre-project loading: 99.8%

6. Facility: KLAMATH CO-GEN / BOURNE 100 KV Line
Contingency: HENRY COUNTY / EMETHUEN SUBSTATION 500 KV
Loading: 113.7% (Rating: 102 MVA)
DFAX: 7.4%
Pre-project loading: 97.8%
7. Facility: SHERMAN ST / KLAMATH CO-GEN SUBSTATION 100 KV Line
Contingency: HENRY COUNTY / E METHUEN 500 KV
Loading: 113.7% (Rating: 109 MVA)
DFAX: 7.4%
Pre-project loading: 98.8%
8. Facility: NORTH TIFTON / W ASHEVILLE 100 KV Line
Contingency: WARTBURG / N TIFTON 100 KV
Loading: 113.0% (Rating: 181 MVA)
DFAX: 12.3%
Pre-project loading: 98.0%
9. Facility: N TIFTON / WARTBURG 100 KV
Contingency: WEST ASHEVILLE / NORTH TIFTON 100 KV
Loading: 112.2% (Rating: 160 MVA)
DFAX: 10.6%
Pre-project loading: 97.6%
10. Facility: OAK PLAINS / EASTMONT 100 KV
Contingency: E METHUEN / HENRY COUNTY 500 KV
Loading: 111.3% (Rating: 133 MVA)
DFAX: 8.0%
Pre-project loading: 98.1%
11. Facility: EASTOWN ROAD / OTTAWA 230 KV Line
Contingency: SWAN VALLEY / E METHUEN 500 KV
Loading: 109.6% (Rating: 448 MVA)
DFAX: 5.6%
Pre-project loading: 106.8%
12. Facility: VICTORY / ADIRONDACK 230 KV
Contingency: E METHUEN / SWAN VALLEY SUBSTATION 500 KV line
Loading: 107.4% (Rating: 488 MVA)
DFAX: 5.6%
Pre-project loading: 104.9%
13. Facility: OTTAWA / VICTORY 230 KV Line
Contingency: E METHUEN / SWAN VLLLEY 500 KV
Loading: 107.2% (Rating: 479 MVA)
DFAX: 5.6%
Pre-project loading: 104.6%
14. Facility: SWA VLY / TWIN COUNTY RANCH 230 KV line
Contingency: SWAN VLY / TREGO 500 KV
Loading: 106.4% (Rating: 773 MVA)
DFAX: 10.9%
Pre-project loading: 103.3%
15. Facility: MONTPELIER ELECTRIC GENERATING STA / DOVILLE 230 KV
Contingency: MONTPELIER ELECTRIC GENERATING STA / WABASH 500 KV
Loading: 102.0% (Rating: 775 MVA)
DFAX: 9.4%
Pre-project loading: 99.3%
16. Facility: WAYNEDALE / MONFORT 230 KV line
Contingency: MONTPELIER ELECTRIC GENERATING STATION / WABASH 500 KV

Loading: 100.5% (Rating: 962 MVA)

DFAX: 8.0%

Pre-project loading: 98.7%

17. Facility: COSO 7 / MONTPELIER ELECTRIC GENERATING STA 100 KV

Contingency: EAST METHUEN / HENRY COUNTY 500 KV

Loading: 100.1% (Rating: 135 MVA)

DFAX: 10.6%

Pre-project loading: 82.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WEKIWA / W FORD FLAT SUBSTATION 100 KV	Rebuild line with higher-capacity conductor	\$9,676,000	100% (\$9,676,000)
COSO 7 / WEST ASHEVILLE 100 KV L	Rebuild line with higher-capacity conductor	\$15,322,000	17% (\$2,566,000)
W ASHEVILLE / WARTBURG 100 KV Line	Rebuild line with higher-capacity conductor	\$6,562,000	18% (\$1,214,000)
RED LION / WEKIWA 100 KV	Rebuild line with higher-capacity conductor	\$6,491,000	100% (\$6,491,000)
SHERMAN ST / EASTMONT 100 KV	Rebuild line with higher-capacity conductor	\$4,292,000	19% (\$829,000)
KLAMATH CO-GEN / BOURNE 100 KV Line	Rebuild line with higher-capacity conductor	\$7,376,000	19% (\$1,424,000)
SHERMAN ST / KLAMATH CO-GEN SUBSTATION 100 KV Line	Rebuild line with higher-capacity conductor	\$8,960,000	19% (\$1,730,000)
NORTH TIFTON / W ASHEVILLE 100 KV Line	Rebuild line with higher-capacity conductor	\$14,224,000	18% (\$2,632,000)
N TIFTON / WARTBURG 100 KV	Rebuild line with higher-capacity conductor	\$4,225,000	18% (\$782,000)
OAK PLAINS / EASTMONT 100 KV	Rebuild line with higher-capacity conductor	\$14,863,000	28% (\$4,195,000)
EASTOWN ROAD / OTTAWA 230 KV Line	Reconductor line	\$7,337,000	6% (\$426,000)
ADIRONDACK / VICTORY 230 KV	Reconductor line	\$15,032,000	6% (\$872,000)
VICTORY / OTTAWA 230 KV line	Reconductor line	\$6,809,000	6% (\$395,000)
SWA VLY / TWIN COUNTY RANCH 230 KV line	Reconductor line	\$12,712,000	5% (\$607,000)
MONTPELIER ELECTRIC GENERATING STATION / DOOVILLE 230 KV	Reconductor line	\$16,735,000	6% (\$984,000)
WAYNE DALE / MONFORT 230 KV line	Reconductor line	\$22,310,000	5% (\$1,092,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
COSO 7 / MONTPELIER ELECTRIC GENERATING STA 100 KV	Reconductor line	\$5,414,000	17% (\$906,000)
	Total allocated to project		\$36,821,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.